

Technical Appendix: The Combinatorial Structure

Explicit Formulas for All Six Pair Types

1 The Difference Quotient Identity

Proposition 1 (PRZZ Lines 1508-1511). *For $N = T^\theta$ and complex α, β with $\text{Re}(\alpha), \text{Re}(\beta) > -1$:*

$$\frac{N^{\alpha x + \beta y} - T^{-\alpha - \beta} N^{-\beta x - \alpha y}}{\alpha + \beta} = N^{\alpha x + \beta y} \log(N^{x+y} T) \int_0^1 (N^{x+y} T)^{-t(\alpha + \beta)} dt \quad (1)$$

At $\alpha = \beta = -R/L$ (where $L = \log T$):

$$(1 + \theta(x + y)) \int_0^1 e^{R[2t + \theta(2t-1)(x+y)]} dt \quad (2)$$

After applying $Q(-\partial/\partial\alpha)Q(-\partial/\partial\beta)$:

$$Q(\theta t(x + y) - \theta y + t) \times Q(\theta t(x + y) - \theta x + t) \times e^{R[\dots]} \quad (3)$$

2 Explicit Pair Formulas

2.1 Pair (1,1): Case A×A

Both pieces have $\omega = -1$, requiring second derivatives.

$$S_{11} = \int_0^1 \int_0^1 (1 - u)^2 \left[\frac{\partial^2}{\partial x \partial y} P_1(x + u) P_1(y + u) \Big|_{x=y=0} \right] Q(t)^2 e^{2Rt} du dt \quad (4)$$

Simplifies to:

$$S_{11} = \int_0^1 \int_0^1 (1 - u)^2 [P_1'(u)]^2 \cdot Q(t)^2 e^{2Rt} du dt \quad (5)$$

2.2 Pair (1,2): Case A×B

Piece 1 has $\omega = -1$ (derivative), piece 2 has $\omega = 0$ (direct).

$$S_{12} = \int_0^1 \int_0^1 (1 - u) \left[\frac{\partial}{\partial x} P_1(x + u) \Big|_{x=0} \right] P_2(u) \cdot Q(t)^2 e^{2Rt} du dt \quad (6)$$

Simplifies to:

$$S_{12} = \int_0^1 \int_0^1 (1 - u) P_1'(u) P_2(u) \cdot Q(t)^2 e^{2Rt} du dt \quad (7)$$

2.3 Pair (1,3): Case A×C

Piece 1 has $\omega = -1$ (derivative), piece 3 has $\omega = +1$ (auxiliary integral).

$$S_{13} = \int_0^1 \int_0^1 \int_0^1 (1-u) P_1'(u) P_3((1-a)u) \cdot a^{\omega_3-1} \cdot Q(\dots)^2 e^{\dots} da du dt \quad (8)$$

For $\omega_3 = 1$: $a^{\omega_3-1} = a^0 = 1$.

The auxiliary integral arises from:

$$\int_{1/q}^1 t^{\alpha+s-1} \log^\tau t dt = \frac{(-1)^\tau \tau!}{(\alpha+s)^{\tau+1}} - \dots \quad (9)$$

2.4 Pair (2,2): Case B×B

Both pieces have $\omega = 0$, no derivatives or auxiliary integrals.

$$S_{22} = \int_0^1 \int_0^1 P_2(u)^2 \cdot Q(t)^2 e^{2Rt} du dt \quad (10)$$

This is the simplest case.

2.5 Pair (2,3): Case B×C

Piece 2 has $\omega = 0$ (direct), piece 3 has $\omega = +1$ (auxiliary integral).

$$S_{23} = \int_0^1 \int_0^1 \int_0^1 P_2(u) P_3((1-a)u) \cdot Q(\dots)^2 e^{\dots} da du dt \quad (11)$$

2.6 Pair (3,3): Case C×C

Both pieces have $\omega = +1$, requiring two auxiliary integrals.

$$S_{33} = \int_0^1 \int_0^1 \int_0^1 \int_0^1 P_3((1-a)u) P_3((1-b)u) \cdot Q(\dots)^2 e^{\dots} da db du dt \quad (12)$$

This is a 4-dimensional integral.

3 Mirror Term Assembly

Proposition 2 (PRZZ Line 1502).

$$I_1(\alpha, \beta) = I_{1,1}(\alpha, \beta) + T^{-\alpha-\beta} I_{1,1}(-\beta, -\alpha) + O(T/L) \quad (13)$$

The complete c formula:

$$c = S_{12}(+R) + m \cdot S_{12}(-R) + S_{34}(+R) \quad (14)$$

where:

$$m = (f_{I_1} g_{I_1} + (1 - f_{I_1}) g_{I_2}) \cdot (e^R + 2K - 1) \quad (15)$$

Correction factors:

$$g_{I_1} = 1 + \frac{\theta(1-\theta)(2(K-1)+\theta)}{8K(2K+1)^2} \quad (16)$$

$$g_{I_2} = 1 + \frac{\theta(2-\theta)}{2K(2K+1)} \quad (17)$$

For $K = 3$, $\theta = 4/7$:

$$g_{I_1} = 1.000952 \quad (18)$$

$$g_{I_2} = 1.019436 \quad (19)$$

$$m \approx 8.814 \quad (20)$$

4 Extension to $K = 4$

For $K = 4$, piece 4 has $\omega(4) = 4 - 2 = 2$, introducing Case D.

Number of pairs: $K(K+1)/2 = 10$

New case types: A×D, B×D, C×D, D×D

Case D involves two auxiliary integrals with $a^{\omega-1} = a^1 = a$ weight.